SPARC: energy-aware production scheduling

Industrial report — shift optimisation, cost simulation and cold-room control at CONCAST

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1. Executive summary

SPARC is a software tool, installed on a plant computer, that moves the flexible production batches within the time window each batch is allowed to move, so that more of their electricity comes from the plant's own cogenerator and solar panels, or is bought when the grid price is lower. It then checks the effect on the electricity bill over any period, and it can simulate a cold room cooled with surplus on-site production.

We tested SPARC on the real CONCAST data: the reference production week, hourly on-site generation from January 2024 to October 2025 and hourly grid prices. Each of the **92 weeks** was optimised with the generation and prices of that week.

Key figures

	Current plants	With FV3 solar plant
Grid electricity cost today (average per year)	€221 100	€188 500
Saving if every week is optimised with its own data	**€4 600/year (2.1%)**	**€5 100/year (2.7%)**
Saving if last week's optimised plan is reused	€1 600/year	€2 000/year
Grid electricity avoided (weekly optimisation)	18 MWh/year	21 MWh/year
Time to optimise one week	about 2 s	about 1 s

What this means.

- The savings are real but modest: only three of the six processes may move (`b_past`, `b_burr` and `b_conf`), they use 16% of the weekly electricity, and they may move by at most two hours.
- The optimisation is exact: for the data it is given, SPARC finds the cheapest possible timetable and proves it (183 of the 184 weekly runs; the last one is within 1.5% of the best).
- How SPARC is used matters more than the algorithm. Reusing one optimised week on the following weeks keeps only a third of the saving, and in about one week out of four it costs more than the original plan. The saving is captured only by re-optimising every week with the expected generation and prices of the coming week.
- The new FV3 solar plant on its own lowers the grid bill by about €32 600 per year with today's timetable; SPARC adds about €5 100 on top and makes better use of the extra midday production.

Recommendations (details in Section 8).

1. Run a **weekly planning routine**: every week, optimise the coming week with the day-ahead prices and a solar/cogeneration forecast, then hand the plan to production.
2. With production, **confirm the rules** the tool does not yet know: order between processes, cleaning time between batches, staff availability. Add them before going live.
3. **Review the flexibility windows**: each extra hour of allowed movement is worth money (Section 4.4).
4. Treat the cold-room module as a **pilot** until its room model is calibrated on measured temperatures.

2. The opportunity

2.1. The plant's electricity today

Table 1 summarises an average week with the reference production timetable. Production needs about 44.5 MWh per week. The cogenerator and the solar panels produce 31.9 MWh, but only 22.8 MWh of it is used by production: the rest (9.1 MWh) is produced when production does not need it, typically at night and at weekends. About half of the electricity is therefore bought from the grid while, at other times, on-site production goes unused.

Average week (Jan 2024 – Oct 2025)	Current plants	With FV3
Electricity used by production	44.5 MWh	44.5 MWh
Produced by the cogenerator	28.5 MWh	28.5 MWh
Produced by the solar panels	3.4 MWh	10.9 MWh
Bought from the grid	21.7 MWh (49%)	18.2 MWh (41%)
On-site production left unused	9.1 MWh	13.1 MWh
Cost of the electricity bought	€ 4 250	€ 3 630

Table 1: Weekly electricity balance with the current production timetable. Unused production is not credited (no sale to the grid is assumed).

2.2. What can move

The production plan is a typical week. Each process has a constant power while it runs and a number of runs (continuous blocks of activity) in the week. Three processes are fixed; the other three may start up to two hours earlier or later (Table 2). `b_past` alone carries 92% of the flexible energy: it is the process that matters.

Process	Power	Hours/week	Runs/week	Energy/week	Allowed movement
`s_finit`	100 kW	119.0	3	11.9 MWh	fixed
`s_torre`	130 kW	129.5	2	16.8 MWh	fixed
`s_hiro`	134 kW	65.3	3	8.7 MWh	fixed
`b_past`	278 kW	23.3	5	6.5 MWh	up to 2 h earlier or later
`b_burr`	5 kW	25.0	9	0.1 MWh	up to 2 h earlier or later
`b_conf`	19 kW	24.8	8	0.5 MWh	up to 2 h earlier or later
Total			30	44.5 MWh	7.1 MWh flexible (16%)

Table 2: Processes of the reference week (timetable of 22–28 September 2025, file `data/production_timetable.csv`; powers and movement from `data/production.yaml`).

Figure 1 shows the principle on one batch: moving it two hours later puts most of it under the midday solar production, so less electricity is bought.

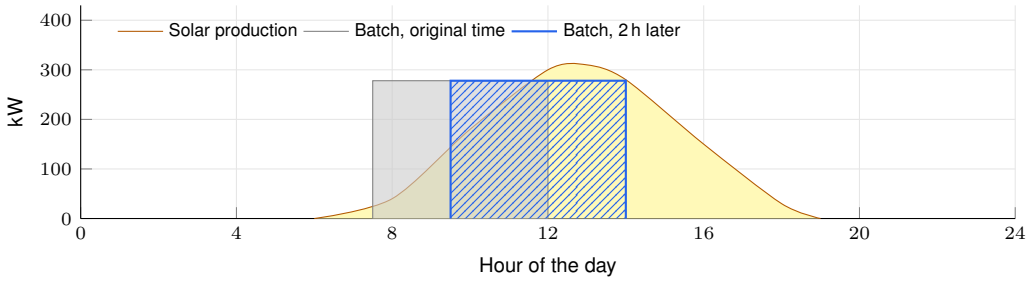


Figure 1: Illustration (not real data): a 278 kW `b_past` batch moved two hours later overlaps much more with solar production.

3. What SPARC does

SPARC has three functions, used through a web browser:

Shift optimisation — for a production plan and a reference week, it chooses the start time of every flexible run so that the electricity bought from the grid in that week costs as little as possible. The result is saved as a new production plan.

Cost simulation — for any period (a week, a month, a season), it calculates the electricity bought and its cost for one plan, or for two plans side by side. This is how an optimised plan is checked before it is adopted.

Cold-room simulation (pilot) — it simulates a cold room kept within its temperature range, cooling first with on-site production and pre-cooling when there is surplus (Section 5).

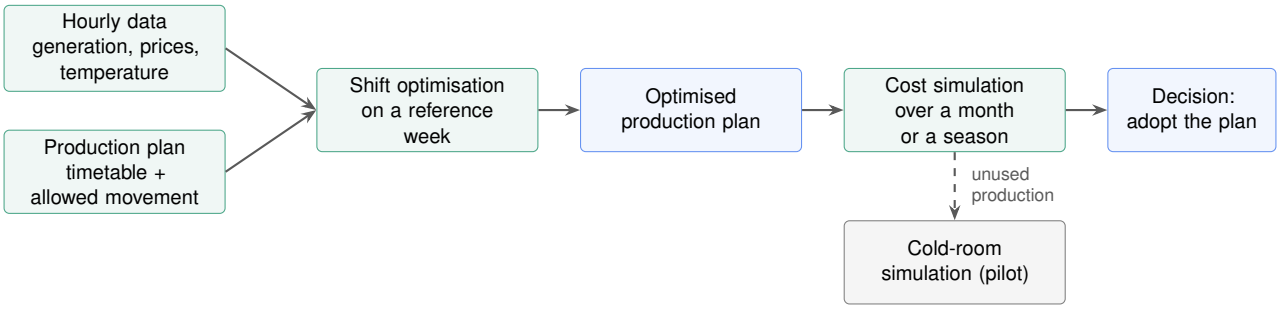


Figure 2: How SPARC is used. Data are entered once; optimisation and checks take seconds.

3.1. How the optimisation decides

The week is divided into 5-minute steps (2 016 per week). For every flexible run SPARC considers every allowed start time — for a run that may move two hours either way, 49 possibilities — and picks one per run so that the cost of the electricity bought from the grid in the week is the lowest. At each 5-minute step, production first uses the on-site generation; only what is missing is bought, at that hour's price.

The schedule always respects these rules:

- a run moves only within its allowed window (fixed runs never move);
- a run keeps its length and power;
- two runs of the same process never overlap (one machine cannot do two batches at once).

Technically this is a mixed-integer linear programme (Appendix A) solved with the open-source solver HiGHS. For one week it has about 1 100 yes/no decisions and is solved to proven optimality in 1–2 seconds; a 20-second limit bounds the wait in rare hard cases, and the page shows the progress of the calculation while it runs.

What the optimisation does not know (yet)

- Order between processes: if one batch must finish before another can start, this is not enforced. Each process is moved on its own.
- Cleaning or changeover time between two runs of the same process: runs may end up back to back.
- Staff, shifts of people, deliveries and storage limits.
- Value of unused production (sale to the grid), power peaks and capacity charges, fuel cost of the cogenerator: only the energy bought from the grid is counted.

A proposed plan must therefore be reviewed by production planning before it is used.

4. Results on CONCAST data

4.1. How the results were obtained

Every full week from Monday 1 January 2024 to Sunday 5 October 2025 (92 weeks) was optimised with the generation and prices of that week, and compared with the original timetable. This was done twice: with the current plants (`generation.csv`) and with the additional FV3 solar plant (`generation_fv3.csv`). The same reference timetable was used for every week. The calculation uses exactly the code of the tool (Appendix B).

4.2. Savings

92 weeks, every week optimised with its own data	Current plants	With FV3
Grid cost, original timetable	€ 391 207	€ 333 499
Grid cost, optimised timetables	€ 383 014	€ 324 449
Saving	**€ 8 193 (2.1%)**	**€ 9 050 (2.7%)**
Saving per week: lowest / typical / highest	€ 28 / € 80 / € 215	€ 33 / € 92 / € 251
Saving per year (52 weeks)	€ 4 630	€ 5 120
Grid electricity avoided per year	18.4 MWh	21.4 MWh
CO_2 avoided per year (0.5 kg/kWh)	9 t	11 t
Flexible runs moved (typical week)	21 of 22	21 of 22

Table 3: Savings when each week is optimised knowing that week's generation and prices. This is the most that can be obtained with the current flexibility windows.

The saving is positive every week, but it varies (Figure 3): it depends on how much on-site production and how much price difference fall within two hours of the flexible runs. In percentage terms it is highest in spring and summer, when solar production is large (Table 4); in winter the bill is high but the flexible runs have little solar production to move into. The grid electricity avoided is exactly on-site production that was previously unused: SPARC does not reduce consumption, it uses the plant's own production better.

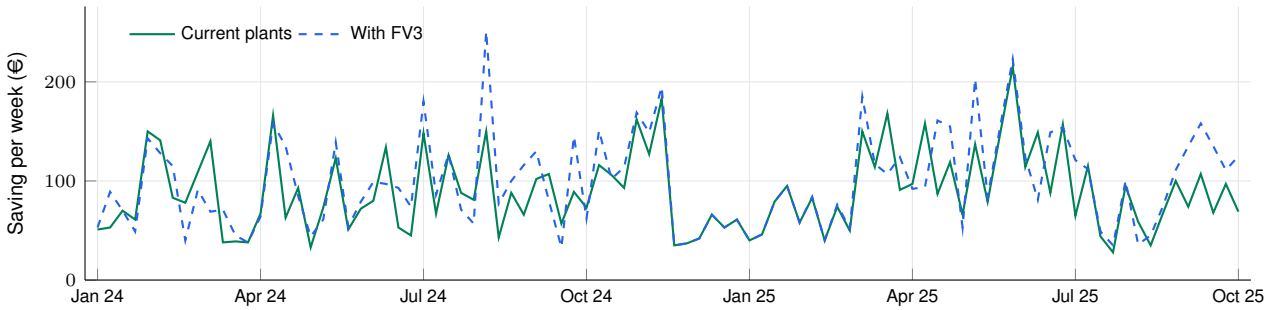


Figure 3: Saving of each week when it is optimised with its own generation and prices.

		Current plants		With FV3	
Season	**Weeks**	**Grid cost/week**	**Saving/week**	**Grid cost/week**	**Saving/week**
Winter (Dec–Feb)	22	€ 7 059	€ 72 (1.0%)	€ 6 525	€ 71 (1.1%)
Spring (Mar–May)	26	€ 2 731	€ 101 (3.7%)	€ 2 184	€ 107 (4.9%)
Summer (Jun–Aug)	26	€ 3 412	€ 88 (2.6%)	€ 2 564	€ 101 (3.9%)
Autumn (Sep–Nov)	18	€ 4 233	€ 95 (2.2%)	€ 3 696	€ 115 (3.1%)

Table 4: Savings by season (weeks grouped by the month in which they start).

4.3. The way SPARC is used decides the saving

The figures above assume that each week is optimised knowing its own generation and prices. In practice the plan is prepared before the week starts. We therefore tested two simple ways of working that need no forecast (Table 5).

Way of working (current plants)	Saving per year	Share of the potential	Weeks costing more than today
Optimise each week with its own data (upper bound)	€ 4 630	100%	none
Reuse last week's optimised plan	€ 1 600	34%	22 of 91
One optimised plan per season, reused all season	€ 1 070	23%	27 of 92
With FV3: optimise each week / reuse last week's plan	€ 5 120 / € 2 000	100% / 38%	none / 17 of 91

Table 5: Saving actually obtained depending on how the optimised plan is prepared. The seasonal plans were optimised on a typical 2024 week of each season.

The reason is simple: the shifts move by at most two hours, so the best position depends on the hour-by-hour pattern of the specific week — whether the cogenerator runs, whether the morning is sunny, when prices peak. A plan tuned on another week moves batches into hours that may be worse in the week where it is applied. The example in the tool shows the same effect: the plan optimised on the week of 15 July 2024 saves 5.2% on that week but less than 1% over the whole of July.

Practical consequence

To capture the saving, SPARC should be run **every week, for the coming week**, with the best available expectation of that week: day-ahead electricity prices (known the day before), a solar production forecast and the planned cogenerator operation. With perfect information the potential is € 4 600–€ 5 100 per year; a good forecast should recover most of it, which is to be verified in the pilot (Section 8).

4.4. The value of more flexibility

The two-hour window is the main limit. We repeated the optimisation on one week in four (23 weeks, current plants), giving the three flexible processes a window of one to four hours each way (Table 6).

Allowed movement of `b_past`, `b_burr`, `b_conf`	Saving	Saving per year	Longest solve
Up to 1 hour earlier or later	€ 1 133 (1.1%)	€ 2 560	1.5 s
Up to 2 hours (today's setting)	€ 1 884 (1.8%)	€ 4 260	7 s
Up to 3 hours	€ 2 552 (2.5%)	€ 5 770	18 s
Up to 4 hours	€ 3 108 (3.0%)*	€ 7 030*	20 s (limit)

Table 6: Saving (each week optimised with its own data) against the width of the window, on 23 weeks spread over the 92. *Some weeks reached the 20-second limit: the true saving is slightly higher.

Each extra hour of allowed movement adds roughly € 1 300–€ 1 700 per year: three hours each way would give about 35% more saving than today, four hours about 65% more. (On this sample of weeks, today's setting gives € 4 260 per year instead of the € 4 630 of the full 92 weeks; the comparison between rows is what matters.) Wider windows make the calculation longer: with four hours some weeks reach the 20-second limit of the web page, which should then be raised.

4.5. Comparison with the earlier method

SPARC also contains the first method developed in the project (a least-squares fit followed by a local search). On the CONCAST data it produced, in every one of the 92 weeks, plans in which two batches of the same process overlap — for example `b_burr` running two batches at the same time on Monday. Such plans cannot be executed, and their apparent savings are not real. The new method never does this and is proven optimal. The earlier method is therefore hidden from users and kept only for developers' comparisons.

5. Cold-room module (pilot)

The cold-room module simulates one cold room over a period, in 5-minute steps:

- the room gains heat through its walls, in proportion to the insulation (U-value), the surface and the difference with the outside temperature (hourly measurements, Trento);
- a controller (proportional–derivative) sets the cooling power to keep the room near its target and inside the allowed range; a minimum runtime avoids frequent on/off cycles;
- cooling uses on-site production first; when there is a surplus, the room is cooled a little further (“stores cold”); grid electricity is used only if allowed;
- it can be linked to a cost simulation, so that the cold room only uses the production left unused by the processes.

It reports the room temperature, the energy used and how much of it came from the grid. It does not consider the price of electricity, and the conversion from heat to temperature uses a fixed calibration coefficient, not the measured behaviour of a real room.

Status

Useful to explore ideas (for example: how much of the 9 MWh/week of unused production a cold room could absorb). Before its figures are used for decisions, the room model must be calibrated against logged temperatures and electricity use of a real CONCAST cold room.

6. Installing and running SPARC

6.1. Installation

SPARC runs on one computer with **Docker Desktop** (Windows, macOS or Linux; its licence is free only below a certain company size, so check it with IT) or, on a Linux server, with the free Docker Engine. The software folder contains two launchers: `start` (double-click) installs everything on the first run, starts SPARC and opens the browser; `stop` stops it. No database, password file or command line is needed.

Operation	Time measured
First installation (download and preparation, once)	a few minutes, depending on the network
First start with an empty database, CONCAST data loaded and examples calculated	9 s
Normal start or restart	2 s
Optimise one week	0.5–2 s
Cost simulation of one month, two plans	0.2 s

Table 7: Times measured on a development laptop with Docker Desktop.

On the first start SPARC already contains the CONCAST data (both generation scenarios, prices 2024–2025, Trento temperatures, the reference week) and one calculated example of each function. The default login (`admin`/`sparc`) must be changed at the first use. By default SPARC can be opened only from the computer where it runs; a setting opens it to the plant network.

6.2. Using it

The browser shows two areas:

- **Results**: a home page explaining the steps, the list of optimisations, cost simulations and cold-room simulations with their charts, and each production week as a chart. Anything not yet calculated has a Calculate now button.
- **Manage data**: forms to upload data and set up calculations, in the order of the work (1 Input data, 2 Production plans and cost simulations, 3 Shift optimisation, 4 Cold rooms). Each calculation has a Save and calculate button and shows its result on the same page. Uploaded files are checked (missing hours, wrong columns) and errors are explained before anything is saved. When a setting changes, the old result is removed so that an outdated figure is never shown.

6.3. Data needed

Data	Resolution	Content and source
On-site generation	hourly, CSV	kWh produced by the cogenerator and by the solar panels (plant metering; for planning: forecast)
Electricity prices	hourly, CSV	price paid per kWh bought (supplier contract or day-ahead market price plus fixed components)
Outside temperature	hourly, CSV	only for cold rooms (weather station or weather service)
Production plan	15 minutes, CSV + YAML	which process runs when in a typical week; power of each process; how far each shift may move

Table 8: Inputs. The file formats are shown on the upload pages and in the README.

6.4. Proposed weekly routine

1. **Thursday**/**Friday**: update the production plan of next week if it differs from the typical week; upload the price and generation expectation for next week.
2. Create a shift optimisation for next week and press Save and optimise (seconds).
3. Production planning reviews the moved batches (list and weekly chart) against the rules SPARC does not know; adjust and re-run if needed.
4. **After the week**: upload the actual generation and prices and run a cost simulation of the plan that was used against the original one, to track the real saving.

7. Limitations and risks

Limitation	Effect	Mitigation
Dependencies between processes, cleaning times, staff are not modelled	A proposed plan may be impossible to run as it is	Review by production planning; add these rules to the model (simple extension)
Savings measured with perfect knowledge of each week	Real savings will be lower, depending on forecast quality	Weekly routine with forecasts; track the real saving after each week
Same typical week used for all 92 weeks	Holidays, maintenance and changes in volume are not reflected	Use the actual plan of each week when available
Only electricity bought is counted	Unused production, power peaks, capacity charges and cogenerator fuel are not valued	Add a value for exported energy and a peak-power charge if relevant to the contract
Power of each process is constant while it runs	Real power profiles within a batch are smoothed	Use measured profiles per process if available
Cold-room model not calibrated	Temperatures and energies are indicative	Calibrate on logged data before use
One computer, local database	Data are lost if the computer is lost	Regular backup of the SPARC data volume

Table 9: Main limitations of the current version.

8. Recommendations and next steps

The actions are in order of priority. The first two are needed before plans proposed by SPARC are used in production; the others increase the saving.

	Action	Expected benefit	Effort
1	**Pilot of the weekly routine** for 8–12 weeks in spring/summer, with day-ahead prices and a solar forecast; record the plan used and the real saving each week.	Measures the share of the €4 600–€5 100/year that is reachable	Low: planner time, data feeds
2	**Workshop with production** to list the rules between processes (order, cleaning, staff) and add them to the optimisation.	Plans that can be used without manual correction	Low–medium
3	**Review the movement windows** of the flexible processes, starting with `b_past`; consider making more processes movable.	See Table 6	Organisational
4	**Value the unused production** (about 9 MWh/week): export price, thermal storage or cold rooms; add it to the cost if it can be sold.	Larger than the shift saving if a use is found	Medium
5	**Calibrate the cold-room model** on one instrumented room, then add price awareness.	Reliable cold-room figures	Medium
6	**Automatic data feeds** for prices and generation, and a scheduled weekly optimisation.	Routine without manual uploads	Medium

Table 10: Recommended actions, in order of priority.

A. Appendix: optimisation model

For engineers who want to check the method. The week is divided into steps $t = 1, \dots, N$ of 5 minutes ($N = 2016$). Each run r has a demand profile $\boldsymbol{d}_r$ (kWh per step) and an activity profile $\boldsymbol{a}_r$ (1 while it runs); T_k shifts a profile by k steps. Run r may move by $k \in \{-B_r, \dots, F_r\}$ (backward and forward limits of its shift). Generation g_t and price $\pi_t \geq 0$ are known. The binary $x_{r,k}$ is 1 when run r is moved by k; $y_t \geq 0$ is the energy bought at step t.

$$
\begin{aligned}
\min_{x,\,y} \quad & \sum_{t=1}^{N} \pi_t \, y_t \\
\text{s.t.} \quad & \sum_{k=-B_r}^{F_r} x_{r,k} = 1 && \text{every run } r \\
& \sum_{r} \sum_{k} x_{r,k} \, (T_k \boldsymbol{d}_r)_t - y_t \leq g_t && \text{every step } t \\
& \sum_{r \in p} \sum_{k} x_{r,k} \, (T_k \boldsymbol{a}_r)_t \leq 1 && \text{every process } p, \text{ step } t \\
& x_{r,k} \in \{0,1\}, \quad y_t \geq 0.
\end{aligned}
$$

The second constraint with $y_t \geq 0$ makes y_t equal to the energy bought, $\max(0, D_t - g_t)$, at the optimum (prices are not negative); the third prevents two runs of the same process from overlapping. The model is solved by HiGHS through `scipy.optimize.milp` with a relative optimality gap of 10^{-4} and a 20-second limit; if no better schedule is found, the original one is kept. Hourly generation is interpolated linearly to 5-minute steps; prices are constant within the hour.

B. Appendix: how the figures were produced

- Data, in the `data` folder of the software:
  - `generation.csv` and `generation_fv3.csv`: 1 January 2024 – 10 October 2025;
  - `prices.csv`: 2024–2025, from 0.076 to 0.365 €/kWh, average 0.175 €/kWh;
  - `production_timetable.csv` and `production.yaml`: reference week and processes.
- For each of the 92 full weeks and each generation file, the optimisation problem was built exactly as the tool builds it (same 5-minute grid, run detection, interpolation) and solved with the optimiser of the tool. Costs are those of the tool's cost function.
- "Reuse last week's plan": the moves found for week $w-1$ applied to week w. "One plan per season": the moves found for the 2024 week of median cost of each season, applied to all weeks of that season in both years.
- Per-year figures are the 92-week (or 91-week) totals divided by the number of weeks and multiplied by 52. CO_2 uses the tool's factor of 0.5 kg per kWh bought.
- Local time is used throughout; the effect of the two daylight-saving changes per year is negligible for these totals.
- The scripts are in `report/analysis` (`benchmark.py`, `strategies.py`, `sensitivity.py`); the technical note on the optimisation explains the method in detail.